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教育背景

人工智能硕士	2025.03 - 2026.09
奥克兰大学 (QS 67) 一等荣誉	
理学学士 (统计) / 商学学士 (金融)	2019.03 - 2024.09
奥克兰大学 (QS 67)	

实习经历

大模型应用开发	2026.03 - 2026.07
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AI 零部件识别与 workflows 自动化, Allied Medical Limited, 奥克兰

- 基于 Java / Spring Boot 搭建 AI 零部件识别后端, 集成 Spring AI、Azure OpenAI GPT-4o、PostgreSQL/pgvector、MongoDB 与 Docker; 使用 @Async + CompletableFuture 实现异步多模态任务处理, 支撑邮件、图片及 PDF 等多类型输入。
- 构建多模态 Hybrid RAG 链路, 对接 5 万+ ERP 库存记录、产品手册、PDF 与图片证据; 依据文档目录与内容板块进行结构化语义切分, 并通过 OneDrive 监听实现文件自动摄入、按数据源刷新及过期向量清理。
- 设计 Top-20 粗召回 → rerank → Top-5 候选的检索与重排流程, 并建立置信度/证据评分机制, 将结果划分为 Evidence (0.80 - 1.00) 与 Provisional (0.55 - 0.79); 低置信度结果进入 HITL 人工确认, 降低模型幻觉与错误确定性输出。
- 集成 Front Webhook / REST API 与异步工作流, 生成可编辑内部评论并保留人工审核闭环; 项目评测达到 Top-1 76.32%、Recall@5 88.16%。

大模型应用与智能体设计	2025.03 - 2025.07
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无人机出口管制咨询智能体, 新西兰海关, 奥克兰

- 梳理 UAV 出口管制业务流程、咨询场景及 NZSGL 规则, 整理法规文档、UAV 术语及相关 corpus 构建知识库, 并将复杂监管逻辑拆解为可执行 Agent 模块。
- 基于 IBM Watsonx 环境搭建出口管制咨询 Agent, 采用 LangGraph + ReAct 组织 Agent 推理与业务流程, 结合 RAG 实现法规检索、术语匹配、结构化输出及人工复核。
- 设计并实现 Listing、Initial Assessment、Catch-All、Exemption、Final Conclusion 五阶段 workflow, 根据前序判断结果进行状态流转、条件分支及法规检索。
- 采用 Zero/Few-Shot prompting, 构建 15+ 边界场景评测集, 验证 Agent 在模糊、临界及军民两用场景下的判断; 最终选用 LLaMA-3-70B-Instruct, 形成可复核的结构化咨询输出及边界场景处理规则。

项目经历

CareTrip Ops 旅伴办智能体平台	2026.07 - 2026.08
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GOAI Agent Infra 新智基座, 杭州

- 面向旅行社、区域 OTA 与文旅集团, 设计银发出行智能客服与交易协同平台, 将模糊语音需求转化为 可比较、可确认、可执行、可核验、可售后 的组合订单, 覆盖需求澄清、方案生成、库存锁定、支付、出票核验与售后闭环。
- 基于 AgentTeams 设计 1 Case Manager + 5 Worker 的 6 Agent 协同架构, 按编排审批、只读认知、写与核验、异步沉淀四类权限域划分职责, 并通过状态机、Checkpoint 与最小上下文传递实现任务拆解、协同执行与全链路追踪。
- 设计 16 个核心 Skills, 按九字段契约定义输入输出、调用条件、依赖工具、失败处理、安全边界与协同关系, 并通过 MCP 对接供应商、订单中台、支付网关和通知服务; 引入 L1-L4 风险分级控制查询、可逆、交易与高危操作。
- 完善工程可靠性与安全治理, 引入幂等、重试、熔断降级、Checkpoint 续跑、双重核验、三账对账、HITL 与可观测性; 设计 9 类沙箱测试场景, 目标包括需求字段准确率 ≥90%、可执行方案率 ≥85%、工具/API 成功率 ≥95%。

多模态情绪感知对话式 AI 系统	2026.03 - 2026.07
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奥克兰大学, 奥克兰

- 构建端到端多模态 AI 系统, 集成实时面部情绪识别、文本与语音交互、FastAPI 后端编排、SQLite 记忆模块、Groq Llama-3.3-70B 对话生成 及人物性格设定。
- 对比 5 种视觉模型架构, 包括 ConvNeXtV2-Huge、Swin-Large、EfficientNetV2-M、MobileNetV3-Large 和 ResNet50, 最终选择 ConvNeXtV2-Huge 部署。
- 使用 AffectNet-HQ、RAF-DB 和 FER2013 训练情绪模型, ConvNeXtV2-Huge 达到 85.69% 验证准确率和 85.34% 测试准确率。
- 实现 YOLOv8n-face、1.3 秒情绪轮询、三次平滑、FasterWhisper、语音通话、TTS、逐轮情绪时间线追踪, 将情绪状态与人物设定用于多轮回复生成。

技术技能

编程: Java, Python, SQL, R

AI / LLM / ML: RAG, Prompt Engineering, LangGraph, ReAct, MCP, GPT-4o, Llama 3.3 70B, IBM Watsonx, PyTorch, Hugging Face Transformers, scikit-learn, YOLOv8, ConvNeXtV2, FasterWhisper

Backend / Data: Spring Boot, Spring AI, FastAPI, Docker, PostgreSQL, pgvector, MongoDB, SQLite, REST API, Webhooks

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EDUCATION

Master of Artificial Intelligence University of Auckland (QS 67) First Class Honours	Mar 2025 – Sep 2026
Bachelor of Science in Statistics / Bachelor of Commerce in Finance University of Auckland (QS 67)	Mar 2019 – Sep 2024

INTERNSHIP EXPERIENCE

LLM Application Development AI Parts Identification & Workflow Automation, Allied Medical Limited, Auckland	Mar 2026 - Jul 2026
- Developed an AI-assisted parts identification backend with Java / Spring Boot, integrating Spring AI, Azure OpenAI GPT-4o, PostgreSQL/pgvector, MongoDB, and Docker; used @Async + CompletableFuture for asynchronous multimodal task processing across email, image, and PDF inputs. - Built a multimodal Hybrid RAG pipeline connecting 50,000+ ERP inventory records, product manuals, PDFs, and image evidence; applied structure-aware semantic chunking based on document sections and content hierarchy, and implemented OneDrive monitoring for automatic ingestion, source-based refresh, and stale-vector cleanup. - Designed a Top-20 coarse retrieval → rerank → Top-5 candidate selection pipeline and a confidence/evidence scoring mechanism, classifying results into Evidence (0.80–1.00) and Provisional (0.55–0.79); routed low-confidence results to HITL review to reduce hallucinations and overconfident incorrect outputs. - Integrated Front Webhook / REST API with asynchronous workflows to generate editable internal comments while retaining manual-review controls; achieved 76.32% Top-1 accuracy and 88.16% Recall@5 in project evaluation.	
Large Language Model Application & Agent Design UAV Export Controls Advisory Agent, New Zealand Customs, Auckland	Mar 2025 - Jul 2025
- Analysed UAV export-control workflows, consultation scenarios, and NZSGL rules; organised regulatory documents, UAV terminology, and related corpora into a knowledge base and translated complex regulatory logic into executable Agent modules. - Built the export-control advisory Agent within the IBM Watsonx environment, using LangGraph + ReAct to orchestrate Agent reasoning and business workflows, with RAG-based regulatory retrieval, terminology matching, structured outputs, and human review. - Designed and implemented a five-stage workflow covering Listing, Initial Assessment, Catch-All, Exemption, and Final Conclusion, with state transitions, conditional branches, and regulatory retrieval driven by upstream decisions. - Applied Zero/Few-Shot prompting and built a 15+ edge-case evaluation set to validate Agent decisions across ambiguous, borderline, and dual-use scenarios; selected LLaMA-3-70B-Instruct, producing auditable structured advisory outputs and edge-case handling rules.	

PROJECT EXPERIENCE

CareTrip Ops Travel Companion Agent Platform GOAI Agent Infra, Hangzhou	Jul 2026 - Aug 2026
- Developed a customer-service and transaction-coordination platform for travel agencies, OTAs, and tourism groups, converting ambiguous voice requests from older travellers into comparable, confirmable, executable, verifiable, and serviceable travel orders across planning, booking, payment, verification, and after-sales workflows. Architected a 6-Agent architecture on AgentTeams, consisting of 1 Case Manager + 5 Workers, with responsibilities separated across orchestration/approval, read-only cognition, write-and-verification, and asynchronous knowledge capture; used state machines, checkpoints, and minimal-context handoff for task decomposition, collaboration, and end-to-end traceability. Defined 16 core Skills and a nine-field contract covering inputs/outputs, invocation conditions, tool dependencies, failure handling, security boundaries, and collaboration relationships; integrated suppliers, order platforms, payment gateways, and notification services through MCP with L1-L4 risk levels for query, reversible, transactional, and high-risk operations. Established engineering reliability and security controls including idempotency, retries, circuit breaking and graceful degradation, checkpoint recovery, dual verification, three-way reconciliation, HITL, and observability; defined 9 sandbox test scenarios with acceptance targets including requirement-field accuracy ≥90%, executable-plan rate ≥85%, and tool/API success rate ≥95%.	
Multimodal Emotion-Aware Conversational AI System University of Auckland, Auckland	Mar 2026 - Jul 2026
- Built an end-to-end multimodal AI system integrating real-time facial emotion recognition, text and voice interaction, FastAPI backend orchestration, SQLite memory, Groq Llama-3.3-70B response generation, and persona configuration. - Compared 5 vision model architectures, including ConvNeXtV2-Huge, Swin-Large, EfficientNetV2-M, MobileNetV3-Large, and ResNet50. - Trained the emotion recognition model using AffectNet-HQ, RAF-DB, and FER2013, with ConvNeXtV2-Huge achieving 85.69% validation accuracy and 85.34% test accuracy. - Implemented YOLOv8n-face, 1.3-second emotion polling, three-result smoothing, FasterWhisper, voice-call mode, TTS, and per-turn emotion timeline tracking, using emotion state and persona context for multi-turn response generation.	

TECHNICAL SKILLS

Programming: Java, Python, SQL, R

AI / LLM / ML: RAG, Prompt Engineering, LangGraph, ReAct, MCP, GPT-4o, Llama 3.3 70B, IBM Watsonx, PyTorch, Hugging Face Transformers, scikit-learn, YOLOv8, ConvNeXtV2, FasterWhisper

Backend / Data: Spring Boot, Spring AI, FastAPI, Docker, PostgreSQL, pgvector, MongoDB, SQLite, REST API, Webhooks